

Multilevel Determinants of Player Market Value in Elite European Football

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1 Introduction

The European football transfer market has evolved from a subjective scouting exercise into a global financial industry exceeding €7 billion annually. For this study, we operationalize market value using crowd-sourced estimates from Transfermarkt, a metric widely validated in sports economics as a robust proxy for actual transfer fees (Herm et al., 2014). While foundational research identifies performance metrics like goals as primary drivers (Frick, 2007), more recent scholarship highlights a “Superstar Effect” where non-performance factors like brand visibility magnify valuation (Franck & Nüesch, 2012; Rosen, 1981). However, the market has recently bifurcated into a pre-2020 hyper-inflationary era, termed the “Neymar Effect” (Poli et al., 2017), and a post-2020 pandemic correction (Drewes et al., 2021). This trajectory was violently interrupted by the COVID-19 pandemic, which caused an estimated €8.7 billion revenue shortfall across European football (Deloitte, 2021). Existing models typically treat players as static units nested within single clubs, ignoring the dynamic reality where players transfer between teams and accumulate value from both their portable human capital and the institutional prestige of their current employer. Neglecting this cross-classified structure likely misestimates the true drivers of market value in this volatile era.

To address these limitations, this study applies a cross-classified multilevel model to a decade of transfer data (2013–2024), centering the analysis at the 2020 pandemic baseline to quantify the impact of this structural break. This approach disentangles individual reputation from club context, allowing us to investigate four critical dynamics: the non-linear “Boom and Bust” market trajectory (RQ1)¹, the biological constraints of the player career arc (RQ2)², the valuation signal of “Squad Status” via relative playing time (RQ3)³, and the existence of contextual premiums for club diversity or discounts for domestic talent in a globalized labor market (RQ4⁴-RQ5⁵).

2 Method

2.1 Sample and Variables

We obtained data from Transfermarkt and UEFA official rankings, both publicly available sources. We used purposive sampling to select the top 21 European clubs based on 10-year averaged UEFA coefficient rankings⁶. We then identified all players who played a full season in any of these clubs from 2013 through 2024 resulting in a sample of 7,023 player-season observations from 2,159 unique players across 21 clubs over 12 seasons. After listwise deletion of cases with missing citizenship data (approximately 2% loss), players averaged 26.5 years of age ($SD = 4.6$). Critically, 441 players (20.4%) played for multiple clubs during the study

¹How has the transfer market evolved relative to the 2020 baseline? Specifically, does the market exhibit a non-linear “Boom and Bust” cycle that linear models miss?

²How does market valuation evolve across a player’s biological career, and at what specific age does this value peak?

³To what degree does “Squad Status”—defined by relative playing time—predict market value independent of raw performance output?

⁴How is club-level international diversity associated with player valuations in a globalized labor market?

⁵How does the market valuation of domestic players compare to that of imported talent when controlling for performance and club context?

⁶UEFA coefficient rankings are a points-based system used by European soccer’s governing body to rank national associations and clubs based on their performance in European competitions, which determines qualification spots and seeding for tournaments like the Champions League and Europa League.

period, justifying our cross-classified modeling approach.

The primary outcome variable is the natural logarithm of the player’s inflation-adjusted market value (in 2024 USD). Predictors are categorized into three distinct domains: (1) Player Characteristics, including age, height, primary position (Attacker, Midfielder, Defender, Goalkeeper), and citizenship status relative to the club (“Homegrown” vs. Foreign); (2) Performance and Status, captured by goals, assists, disciplinary records (red/yellow cards), and playing time; and (3) Club Context, specifically the proportion of foreign players on the roster and the club’s UEFA coefficient, which serves as a proxy for recent competitive success and prestige. For further descriptive statistics on each variable see Appendix A, Table1.

2.2 Cross-Classified Data Structure

Our data exhibit a cross-classified structure where players (i) are constant units, but club membership (j) changes over time (t). When a player transfers from club j_1 to club j_2 , club-level predictors update to reflect the new club’s characteristics. This structure requires modeling player and club as crossed random effects rather than nested effects. Figure 1 is a visual of a cross-classified model example of three players that either move through clubs or remain in the same club throughout their career.

Figure 1. Cross-classified Data Structure Highlights How Players Move Between Clubs Ov



2.3 Model Specification

We specify a cross-classified multilevel model to partition the variance in market value across observations (t), players (i), and clubs (j). Crucially, because club attributes like internationalization and UEFA ranking vary year-to-year, they are modeled as Level 1 (time-varying) predictors rather than static Level 2 traits.

Level 1: Observation Model (Time-Varying Effects)

At the observation level, the log-transformed market value is modeled as a function of the player-club intercept (π_{0ij}), algebraic time trends, biological aging, and dynamic contextual metrics:

$$\begin{aligned}\log(\text{MV})_{tij} = & \pi_{0ij} \\ & + \pi_{1j}(\text{Year}_t) + \pi_{2j}(\text{Year}_t^2) + \pi_{3j}(\text{Year}_t^3) \\ & + \pi_{4j}(\text{Age}_{it} - \bar{A}) + \pi_{5j}(\text{Age}_{it} - \bar{A})^2 \\ & + \pi_{6j}(\text{Mins}_{Z,it}) \\ & + \pi_{7j}(\text{Foreign}_{tj}) + \pi_{8j}(\text{UEFA}_{tj}) \\ & + \text{perf} + \epsilon_{tij}\end{aligned}$$

Note: $MV = \text{Market Value}$. The subscripts tj for Foreign and UEFA denote that these club attributes vary by time t .

Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept (π_{0ij}) is decomposed into a fixed component based on static player characteristics and a random player effect (u_{0i}):

$$\pi_{0ij} = \beta_{0j} + \gamma_1(\text{Height}_i) + \gamma_2(\text{Citizen}_{ij}) + \text{demog} + u_{0i}$$

Level 2b: Club Model (Contextual Trajectories)

The club baseline (β_{0j}) and the time-varying slope parameters (π_{1j}, π_{2j}) include random effects to allow for heterogeneity in both growth velocity and trajectory curvature:

$$\beta_{0j} = \delta_{00} + v_{0j} \quad (\text{Baseline Prestige})$$

$$\pi_{1j} = \delta_{10} + v_{1j} \quad (\text{Linear Growth Deviation})$$

$$\pi_{2j} = \delta_{20} + v_{2j} \quad (\text{Quadratic Curvature Deviation})$$

$$\pi_3 = \delta_{30} \quad (\text{Fixed Cubic Trend})$$

Variance-Covariance Structure

$$u_{0i} \sim N(0, \sigma_u^2) \quad , \quad \begin{pmatrix} v_{0j} \\ v_{1j} \\ v_{2j} \end{pmatrix} \sim N \left(\mathbf{0}, \begin{pmatrix} \sigma_{Int}^2 & \sigma_{01} & \sigma_{02} \\ \sigma_{01} & \sigma_{Lin}^2 & \sigma_{12} \\ \sigma_{02} & \sigma_{12} & \sigma_{Quad}^2 \end{pmatrix} \right)$$

2.4 Variables and Centering Strategies

To facilitate model convergence and ensure substantively meaningful intercepts, we applied specific centering and transformation strategies. Most notably, **Season Year** was centered at the 2020 pandemic baseline (2020=0), allowing us to interpret the intercept as the expected market value at the onset of the COVID-19 structural break. We introduced polynomial terms to capture theorized non-linear dynamics (see Figure 2): a cubic term for time to model the “Boom-and-Bust” market cycle (pre-pandemic inflation followed by post-pandemic deceleration), and a quadratic term for **Age** to empirically test the biological “Inverted-U” career trajectory. Additionally, **Minutes Played** was standardized (Z-scored) relative to each specific club-year to isolate a player’s hierarchical status (e.g., Starter vs. Bench) independent of team-level rotation policies, while performance metrics were centered by position to account for role-specific expectations.

Figure 2. Quadratic vs. Cubic Polynomial Fits for Market Value Over Time



2.5 Model Building and Statistical Software

We employed sequential model building: (1) null model for ICC estimation; (2) linear time polynomial; (3) cubic time polynomial; (4) random slope for quadratic time at club level⁷; (5) player/club-season Level 1 predictors with quadratic age. Model selection was guided by likelihood ratio tests and information criteria. Model selection and comparison was guided by Likelihood Ratio Tests (LRT) to evaluate the statistical significance of nested model improvements. To balance goodness-of-fit against model complexity, we also consulted the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Models yielding significant LRT results ($p < .05$) and lower values on both information criteria were retained for subsequent analysis. We describe each model in Appendix 2, Table 2 alongside ICC, AIC, and BIC values.

All analyses were conducted in R version 4.5.1 using the `lme4` 1.1-37 package for mixed-

⁷We excluded player-level random slopes due to data sparsity; a majority of players appear in the dataset for only one or two seasons, making it mathematically impossible to disentangle their individual growth trajectories from residual error. In contrast, the club-level data provides a robust longitudinal structure with continuous observations across the decade, allowing for the stable estimation of club-specific market value trajectories.

effects model estimation with a Restricted Maximum Likelihood (REML) method. REML was selected to provide unbiased estimates of the variance components by accounting for the degrees of freedom lost when estimating fixed effects. To ensure robust model convergence given the complexity of the random effects structure, we utilized the BOBYQA optimizer, which handles boundary constraints more effectively than the default Nelder-Mead optimizer. Significance tests used Satterthwaite approximations via the `lmerTest` 3.1-3 package.

3 Results

The unconditional means (null) model revealed substantial clustering at the player level. Player-level variance accounted for 78% of the total variance ($ICC_{player} = .75$), while club-level variance accounted for 5% ($ICC_{club} = .03$), highlighting that persistent individual heterogeneity dominates market valuations. Regarding temporal dynamics, model comparisons favored a cubic fixed effect for season year over a linear specification, $\chi^2(2) = 45.66, p < .001$. We subsequently tested for club-specific heterogeneity in growth trajectories. A likelihood ratio test using a 50:50 mixture of χ^2_1 and χ^2_2 distributions⁸ confirmed that adding a random linear slope significantly improved model fit compared to the random-intercept model, $\chi^2(5) = 97.13, p < .001$, indicating that clubs differ significantly not only in their growth rates but also in the curvature of their market value trajectories.⁹ Consequently, our final best-fitting model incorporates these complex random effects alongside time-varying player and club predictors ($\chi^2(19) = 5626, p < .001$ compared to random slopes model). Model comparisons are shown in Appendix C, Table 1. Here we output our final model’s variance components and fixed effects.

⁸We test whether a random slope with quadratic season year at the club level has better fit to the data. We find evidence supporting the need to include these random slopes using a likelihood ratio test with an alternative hypothesis, $H_A = \tau_{11} > 0$ since variance should be larger than 0. To test this hypothesis we use the likelihood ratio test (LRT) simply by $-2[\log \text{likelihood}_{reduced} - \log \text{likelihood}_{full}]$ to examine the difference. This code can be found in Appendix B.

⁹While we included random linear and quadratic slopes, a random cubic slope was excluded due to over-parameterization. Estimating the required 10 variance parameters with only 21 clubs led to singularity; therefore, the quadratic specification represents the maximal complexity supported by the data.

3.1 Model Diagnostics

The model converged successfully with no warnings. Q-Q plots confirmed that player-level random intercepts ($n = 2,159$) adhered well to normality, while club-level random effects ($n = 21$) showed minor tail departures expected with few Level 2 units (Maas & Hox, 2004). The residual versus fitted values plot revealed homoscedastic variance with no systematic patterns. Full diagnostic plots are presented in Appendix C.

Table 1. Estimated Variance Components for Final Model

Component	Variance	SD
Player (Reputation)	0.9254	0.9620
Club (Baseline Prestige)	0.0541	0.2326
Club (Linear Growth)	0.0105	0.1025
Club (Quadratic Curvature)	0.0073	0.0855
Residual (Unexplained)	0.2070	0.4550

Table 2. Fixed Effects Estimates from Final Model

Predictor		SE	t	% Change	
Year (linear)	0.002	0.030	0.08	0.2	
Year ²	-0.270	0.024	-11.05	-23.7	***
Year ³	-0.117	0.011	-10.98	-11.0	***
Height	0.010	0.004	2.81	1.0	**
Defender	-0.403	0.058	-6.97	-33.2	***
Goalkeeper	-1.059	0.095	-11.17	-65.3	***
Midfielder	-0.190	0.058	-3.29	-17.3	**
Domestic citizen	-0.185	0.031	-5.89	-16.9	***
Age (linear)	0.019	0.004	5.02	1.9	***
Age ²	-0.024	0.000	-63.08	-2.4	***
goals	0.005	0.002	2.36	0.5	*
assists	0.010	0.003	3.51	1.0	***

Red cards	0.009	0.022	0.42	0.9	
Yellow cards	-0.002	0.003	-0.49	-0.2	
Minutes (Z)	0.350	0.013	26.20	41.9	***
UEFA Coefficient	0.000	0.001	-0.39	0.0	
Proportion of Foregin Players	0.103	0.104	0.99	10.8	

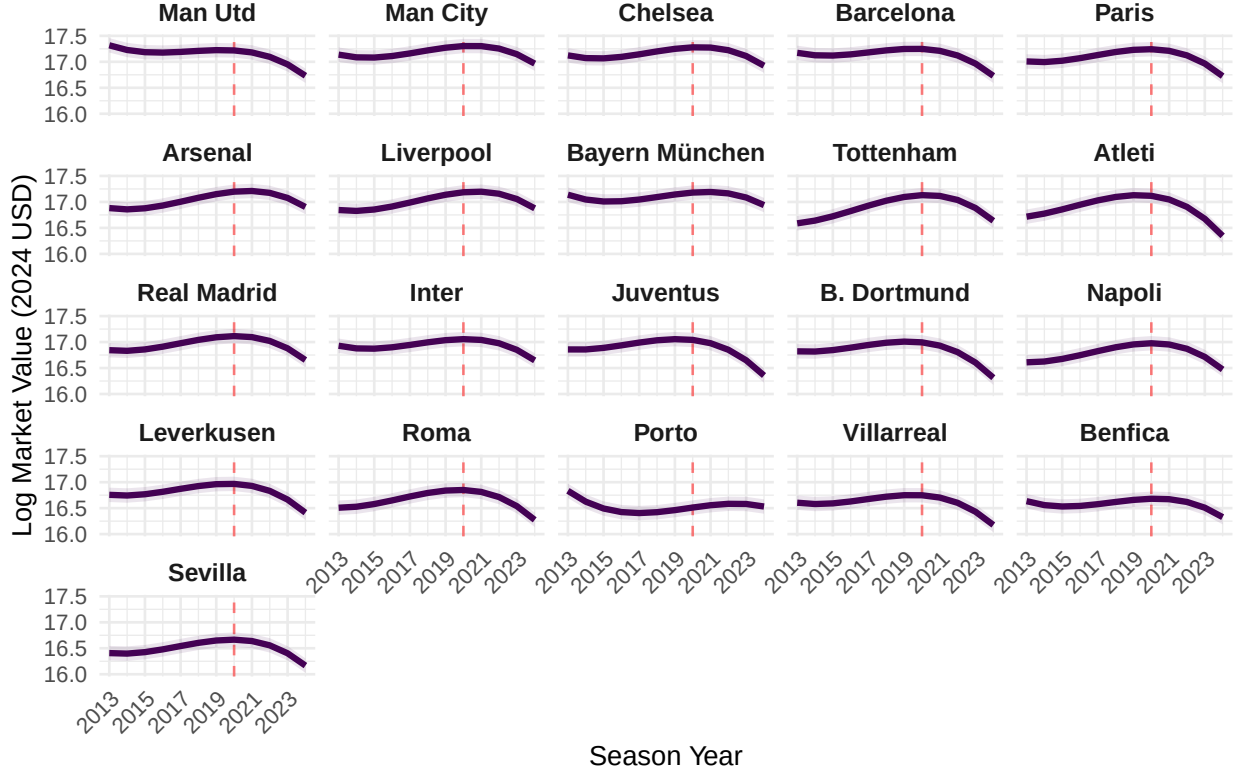
Note: *** $p < .001$, ** $p < .01$, * $p < .05$

3.1.1 Market Evolution

Using raw polynomials centered at 2020 ($t = 0$), our model mathematically pinpoints the moment the market froze. The linear coefficient at the 2020 baseline is effectively zero ($\pi_1 = 0.00, p = 0.93$), indicating that at the exact onset of the pandemic, the relentless inflation of the previous decade came to a complete halt. The significant negative quadratic ($\pi_2 = -0.27$) and cubic ($\pi_3 = -0.12$) terms confirm the transition from a pre-2019 “Boom” into a structural correction. Figure 3 visualizes the complexity of club-level market dynamics by plotting the predicted trajectory for all clubs individually. This faceted view highlights how clubs diverge from the global trend. The vertical position of each chart confirms the initial hierarchy (e.g., random Intercepts), with elite clubs like Manchester City and Bayern Munich maintaining a consistently higher baseline than selling clubs (e.g., Sevilla, Porto). The distinct curvature of each line, particularly around the 2020 pandemic baseline, illustrates the significant variance in the random slopes. Clubs with concave trajectories (e.g., Juventus, Barcelona) plateaued or declined rapidly, confirming the risk of stagnation. Conversely, clubs with convex trajectories (e.g., Arsenal, Liverpool) show stronger momentum, indicating a more resilient recovery and active value generation throughout the decade.

Figure 3. Club-Specific Market Value Trajectories (2013–2024)

Dashed line marks pandemic onset; panels ordered by baseline prestige.

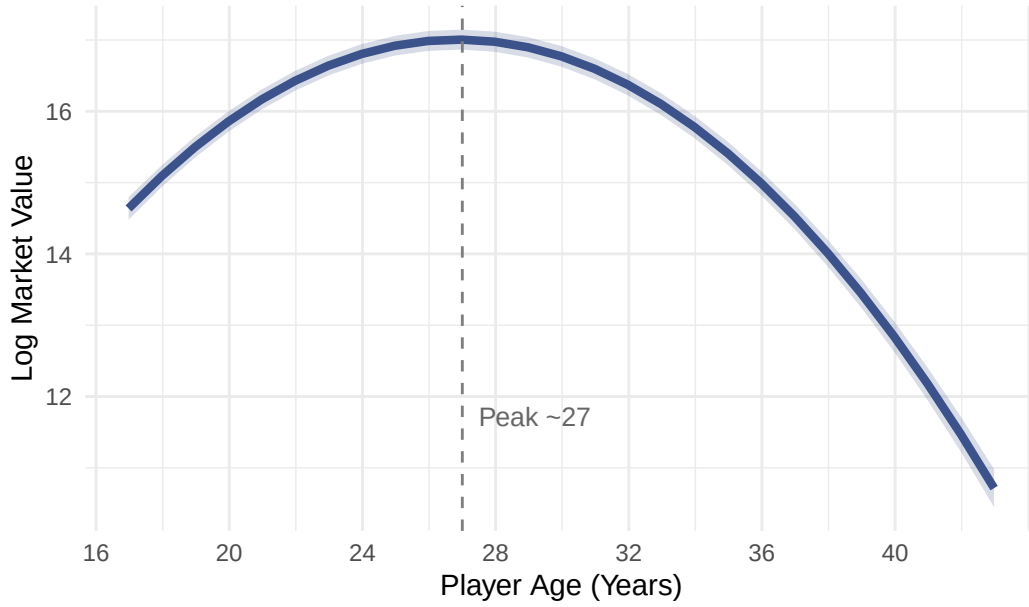


3.1.2 Biological Age

Our model predicts the peak age to be 26.9¹⁰ for optimal transfer market value we visualize in Figure 4 the biological constraints on valuation. The curve rises steeply as players transition from ‘prospects’ (Age 17–23), flattens into a ‘prime’ plateau between ages 26 and 28, and declines steadily thereafter. The model predicts that a player’s value drops by approximately 1.5 log points (from a peak of 17.0 to 15.5) as they age from 27 to 35, reflecting the market’s heavy discount on older assets with limited resale potential.

¹⁰We use the delta method, specifically Taylor series approximation, to propagate uncertainty through nonlinear transformation of $-\beta_{age}/(2 \times \beta_{age}^2)$.

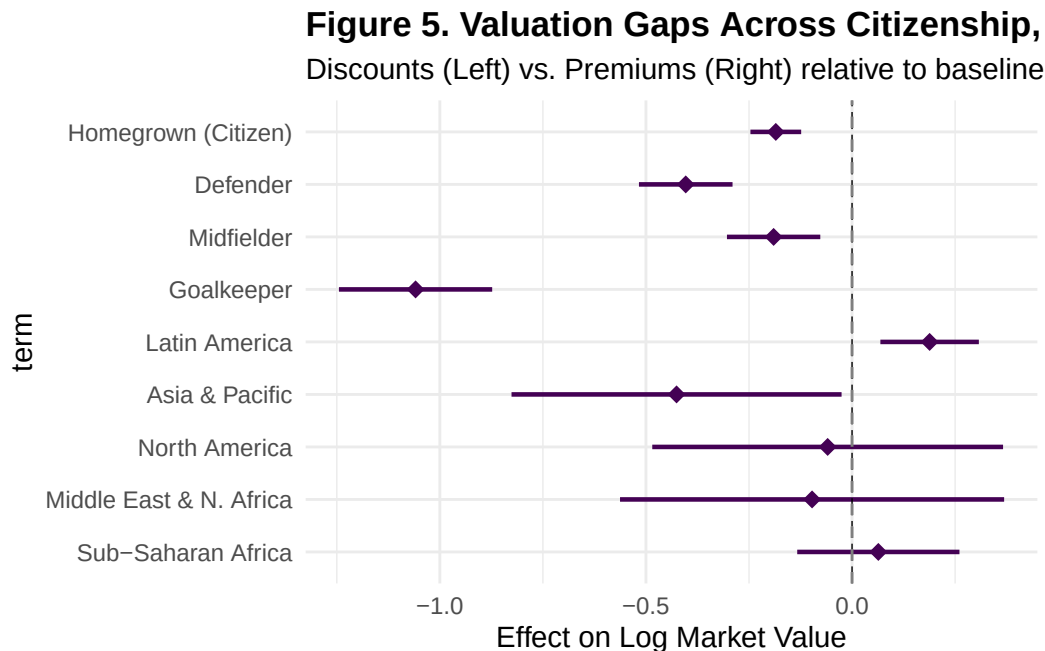
Figure 4. The Career Arc, Predicted Log Market Value by Age



3.1.3 Contextual Premiums and Discounts

Squad status remains the dominant behavioral driver. The standardized coefficient for minutes played ($\pi_6 = 0.35$) indicates a 42% valuation premium ($e^{0.35} - 1$) for key starters (+1 SD) compared to average players. Conversely, we observe a robust “Domestic Discount,” where homegrown citizens are valued approximately 17% lower ($\gamma_2 = -0.19$) than comparable foreign imports. While we hypothesized a “Diversity Premium” for clubs with higher international representation, the fixed effect for `foreign_players` was positive but not statistically significant ($\pi_7 = 0.10, p = 0.32$) in the final model. Similarly, the `uefa_coefficient` showed no significant effect ($p = 0.70$). This suggests that once we account for the massive heterogeneity in Club Random Effects (i.e., Baseline Prestige and Growth Trajectories) and individual player reputation ($ICC \approx 0.78$), specific club-level metrics add little unique explanatory power. In Figure 5, we present relative market valuation for Level 2 fixed effects (e.g., citizenship, region origin, and position). The coefficient for “Homegrown (Citizen)” falls strictly to the left of zero, confirming that domestic players are valued significantly lower than comparable foreign imports ($\gamma_2 = -0.19$), potentially reflecting the recruitment pre-

mium required to attract international talent. Attackers (the baseline) command the highest market value. All other positions show significant negative effects, with Goalkeepers facing the steepest penalty ($\gamma_{pos} = -1.06$), suggesting the market disproportionately rewards offensive output. While most regional coefficients overlap with zero (indicating no difference from the European baseline), Latin America stands out as the only region with a positive premium ($\gamma_{region} = 0.19$), suggesting South American players function as a “premium export brand” in the global market.



4 Discussion

This study demonstrates that player market value in elite football is driven primarily by persistent individual reputation ($ICC = 0.78$) but is meaningfully moderated by club context and market cycles. Our cross-classified approach properly attributes variance to portable player effects versus transient club effects.

Centering our temporal polynomial at 2020 provides direct evidence that the COVID-19 pandemic represented a fundamental discontinuity in the transfer market. The near-zero linear

slope at this inflection point indicates that the decade-long inflationary trend abruptly halted, followed by market correction. The strict inverted-U relationship between age and market value, peaking at approximately 27 years, confirms that transfer markets price both current ability and future option value. The steepness of the post-peak decline (approximately 1.5 log points from age 27 to 35) reflects forward-looking assessment of remaining productive years.

The dominance of minutes played ($\pi_6 = 0.35$) over position-specific metrics suggests that markets rely heavily on managerial assessments as signals of underlying quality, consistent with signaling theory. A one standard deviation increase in relative playing time yields a 42% valuation premium. The finding that homegrown players are valued approximately 17% lower than comparable foreign imports ($\gamma_2 = -0.19$) suggests domestic players may be systematically undervalued while international recruitment requires competitive bidding that inflates prices.

In conclusion, the substantial player-level variance confirms that persistent individual reputation dominates market assessments, while club context creates meaningful premiums and growth trajectories. The transfer market incorporates forward-looking assessments of biological decline, managerial signals through squad selection, and contextual factors that create systematic premiums and discounts. These patterns represent both economic regularities that clubs can anticipate and potential inefficiencies that astute market participants can exploit.

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5 Appendix A

Table 1: Variable Descriptions and Descriptive Statistics ($N=7,168$ raw)

Description	Raw Stats (Mean/SD)	Transformation Strategy
Continuous Variables		
Market Value (2024 USD)	\$25.8M (29.0M)	<i>Log-transformed (ln)</i>
Season Year	2018.63 (3.47)	<i>Centered at 2020 (0 = 2020)</i>
Player Age (Years)	26.50 (4.58)	<i>Grand-mean centered + Quadratic</i>
Height (cm)	182.10 (6.84)	<i>Grand-mean centered</i>
Minutes Played	1761.77 (1337.70)	<i>Z-scored within Club-Year</i>
Goals Scored	3.41 (6.25)	<i>Centered by Position Mean</i>
Assists Made	2.79 (4.00)	<i>Centered by Position Mean</i>
Club Foreign %	63.5% (11.1%)	<i>Grand-mean centered</i>
UEFA Coefficient	20.42 (6.89)	<i>Year-mean centered</i>
Red Cards	0.07 (0.28)	<i>Raw count</i>
Yellow Cards	3.42 (3.60)	<i>Raw count</i>
Categorical Variables		
Citizenship (Homegrown)		<i>Binary (1=Citizen)</i>
- No	4498 (62.8%)	
- Yes	2567 (35.8%)	
- NA	103 (1.4%)	
Primary Position		<i>Factor (Ref: Attack)</i>
- Attack	1925 (26.9%)	
- Defender	2448 (34.2%)	
- Goalkeeper	605 (8.4%)	
- Midfield	2190 (30.6%)	
Region of Origin		<i>Factor (Ref: Europe)</i>
- East Asia & Pacific	69 (1.0%)	
- Europe & Central Asia	5139 (71.7%)	
- Latin America &	1441 (20.1%)	
Caribbean		
- Middle East & North	56 (0.8%)	
Africa		
- North America	52 (0.7%)	
- Sub-Saharan Africa	303 (4.2%)	
- NA	108 (1.5%)	

6 Appendix B

6.1 Annotated Model Syntax

```
#-----  
# MODEL SPECIFICATION & FITTING  
#-----  
  
# 1. MODEL NULL (Unconditional Means)  
# Purpose: Partition variance (calculate ICC) between Player and Club levels.  
model_null <- lmer(usd_revenue_2024_log ~ (1|player_id) + (1|club_id),  
                  data = clean_player_df, REML = TRUE, control = strict_control)  
  
# 2. MODEL LINEAR TIME TREND  
# Purpose: Test for general linear inflation in market values  
## (Base model for time).  
model_time_linear <- lmer(usd_revenue_2024_log ~ season_year_c +  
                          (1|player_id) + (1|club_id),  
                          data = clean_player_df, REML = TRUE, control = strict_control)  
  
# 3. MODEL CUBIC TIME TREND (Fixed Effects)  
# Purpose: Test for non-linear "Boom and Bust" cycles.  
## (raw=TRUE for interpretability)  
model_time_cubic <- lmer(usd_revenue_2024_log ~ poly(season_year_c, 3, raw=TRUE) +  
                          (1|player_id) + (1|club_id),  
                          data = clean_player_df, REML = TRUE, control = strict_control)
```



```

# 4. MODEL RANDOM QUADRATIC SLOPE (Club Level)
# Purpose: Test if trajectory curvature varies by club.
## (Linear slope model omitted for brevity)
model_slope_quadratic <- lmer(usd_revenue_2024_log ~ poly(season_year_c, 3, raw=TRUE) +
                              (1|player_id) + (poly(season_year_c, 2, raw=TRUE) | club_id),
                              data = clean_player_df, REML = TRUE, control = strict_control)

# 5. MODEL (FINAL): FULL SPECIFICATION
# Purpose: Estimate marginal effects of performance, status,
## and context controlling for time.
model_player <- lmer(
  usd_revenue_2024_log ~
    poly(season_year_c, 3, raw=TRUE) +          # Temporal Controls
    poly(age_c, 2, raw=TRUE) +                  # Career Arc
    height_c + position + Region +              # Player Traits
    citizen_of_club_country +                   # Context
    goals + assists + red_cards + yellow_cards + minutes_played + # Performance
    uefa_coefficient + foreign_players +         # club fixed effects
    (1|player_id) + (poly(season_year_c, 2, raw=TRUE) | club_id), # Random Effects
  data = clean_player_df, REML = TRUE, control = strict_control
)

#-----
# LIKELIHOOD RATIO TEST (MIXTURE DISTRIBUTION)
#-----
# Corrects p-values for testing variance components on the boundary (Stram & Lee, 1994)
# LRT test for 2 random slopes

```

```

calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced = as.numeric(logLik(reduced_model))
  ll_full = as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  lrt_stat = -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  df_reduced = attr(logLik(reduced_model), "df")
  df_full = attr(logLik(full_model), "df")
  df_diff = df_full - df_reduced

  # 4. Calculate P-Value based on the difference

  if (df_diff == 3) {
    # --- NEW BLOCK FOR QUADRATIC SLOPE ---
    # Going from 2 Random Effects (Int+Lin) -> 3 Random Effects (Int+Lin+Quad)
    # We add 1 variance + 2 covariances = 3 parameters.
    # The mixture is 0.5 * ChiSq(2) + 0.5 * ChiSq(3)
    p_val = 0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 3, lower.tail = FALSE)

    msg = "Mixture of ChiSq(2) and ChiSq(3)"

  } else if (df_diff == 2) {

```

```

# Going from 1 Random Effect (Int) -> 2 Random Effects (Int+Lin)
# Adds 1 variance + 1 covariance = 2 parameters
p_val = 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
  0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

msg = "Mixture of ChiSq(1) and ChiSq(2)"

} else if (df_diff == 1) {
  # Adding a single parameter (e.g. just a variance, no covariance)
  p_val = 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

  msg = "Mixture of ChiSq(0) and ChiSq(1)"

} else {
  # Fallback for standard fixed effects comparisons
  p_val = pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
  msg = paste0("Standard ChiSq(", df_diff, ")")
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), "\n")
cat("Parameter Diff:", df_diff, "\n")
cat("Distribution: ", msg, "\n")
cat("P-Value:      ", format.pval(p_val, eps = .0001), "\n")
}

```

LRT Statistic: 97.1306

Parameter Diff: 5

Distribution: Standard ChiSq(5)

P-Value: < 0.0001

7 Appendix C

Table 1. Model Fit Comparison (AIC, BIC, and ICC)

Models	Purpose & Research Justification	ICC _{player/club}	AIC	BIC
Null	Calculate ICC and partition variance between players and clubs before adding predictors	.75/.03	20105.19	20132.62
Time Linear	Test for a general linear trend in market value growth	.76/.03	20050.17	20084.46
Time Cubic	Test for non-linear market dynamics (e.g., acceleration or deceleration/plateaus)	.76/.03	20021.61	20069.61
Random Slope Quadratic	Test if the growth curvature varies by club	.76/.04	19934.48	20016.76
Varying Level 1 fixed effects	Estimate the marginal value of performance metrics and player/club characteristics controlling for time	.78/.05	14467.31	14679.87
Final Model Selection	Compare models using LRT, AIC/BIC, and residual diagnostics (QQ plots) to select the most parsimonious and robust fit			

Figure 6. Model Diagnostic Plots

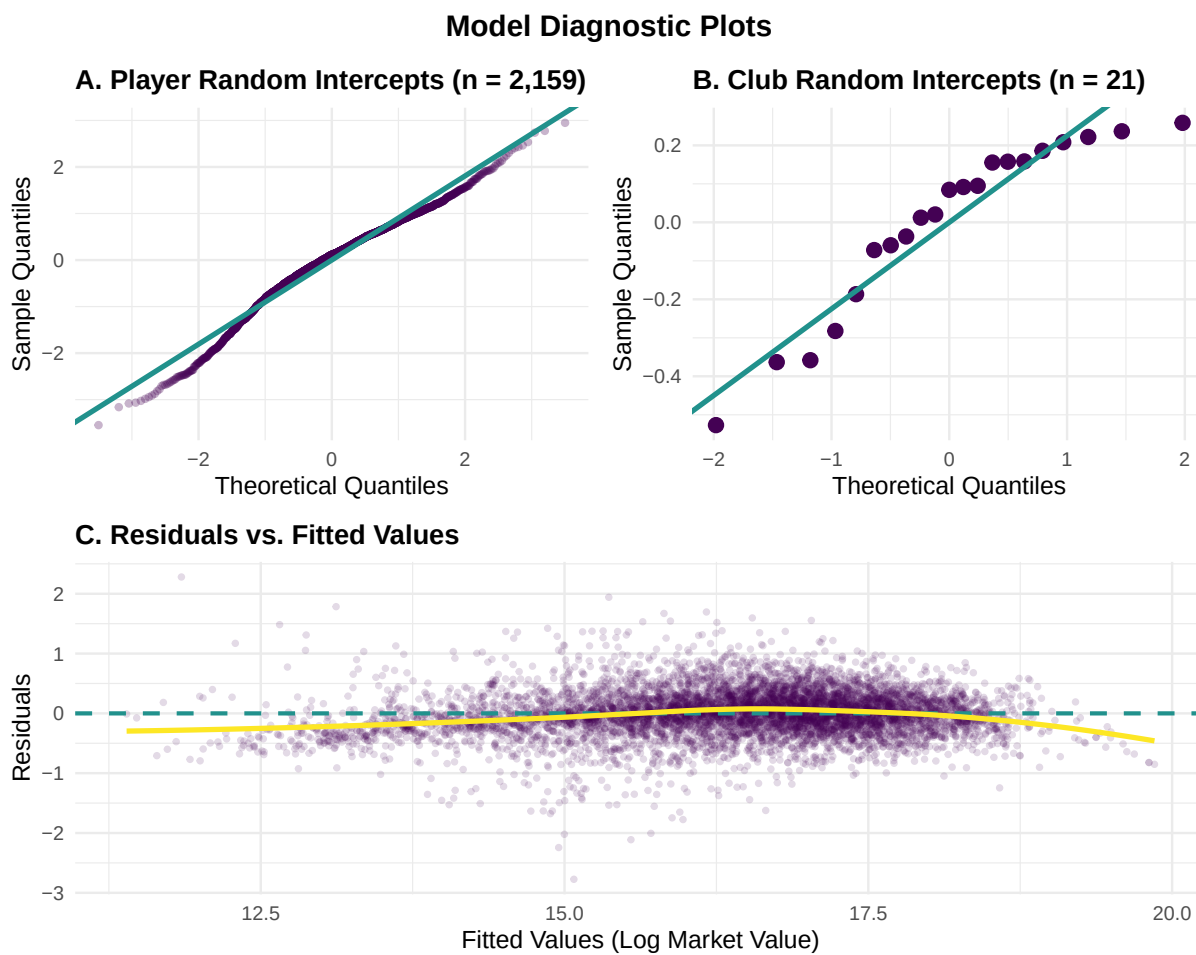


Figure 1: Diagnostic plots: (A) Q-Q plot for player random intercepts, (B) Q-Q plot for club random intercepts, (C) residuals versus fitted values.

Figure 7. Club Random Effects Correlation Structure

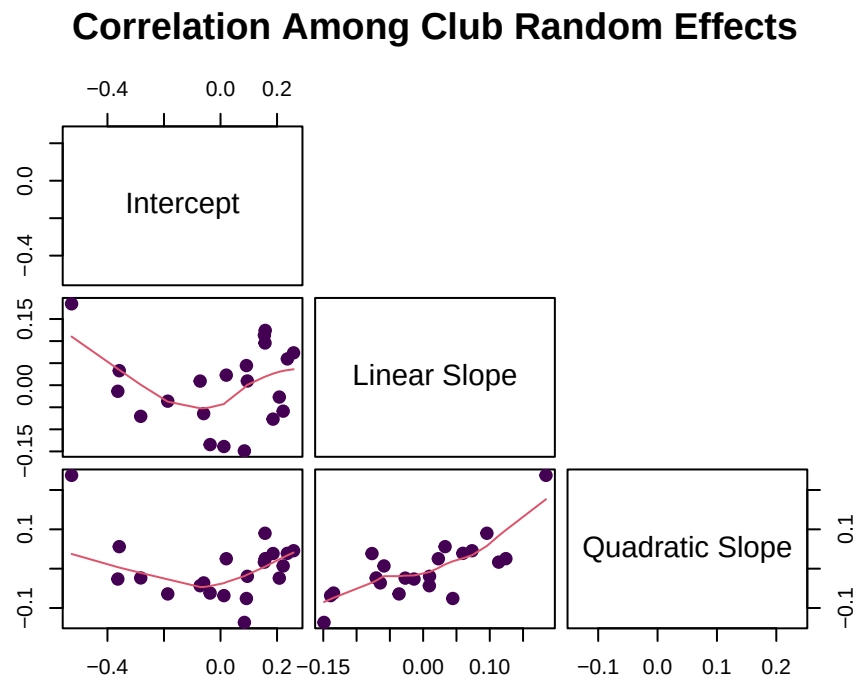


Figure 2: Correlations among club-level random effects (intercept, linear slope, quadratic slope).

8 Appendix D

Acknowledgement of AI Assistance

We acknowledge the use of AI-assisted tools (GitHub Copilot and Claude) during the preparation of this manuscript. These tools were used for:

1. **Coding assistance:** Generating and debugging R code for data visualization, particularly for creating the faceted club trajectory plots (Figure 3) and diagnostic visualizations (Figure 6).
2. **Resolving convergence issues:** Troubleshooting model convergence problems with the cross-classified multilevel model, including optimizer selection (BOBYQA) and parameter constraints.
3. **Code documentation:** Improving comments and annotations in the model syntax presented in Appendix B.

All statistical analyses, interpretations, and substantive conclusions were developed and verified by the authors. The AI tools served as coding assistants and did not contribute to the research design, data collection, or theoretical framework of this study.